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ReRAPTOR: Reverse-Objective RAPTOR for Time-Budgeted Value-Maximizing Transit Itinerary Planning

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Abstract Public-transit routing is usually formulated as an earliest-arrival problem, but leisure exploration and time-budgeted outing design require timetable-feasible routes that maximize itinerary value rather than merely minimize travel time. This paper studies time-budgeted value-maximizing transit itinerary planning on timetable-based transit schedules and proposes ReRAPTOR, a reverse-objective RAPTOR (Round-based Public Transit Optimized Router)-style label-setting method that preserves timetable-feasible expansion while ranking partial routes by a value-aware surrogate score. The framework supports pluggable objective modules and uses three pruning layers: Beam pruning for width control, Dominance pruning for local label filtering, and Negative pruning for suppressing semantically weak partial routes. We evaluate the method on transit schedule data for Fukuoka provided by Ekispert API (`https://api-info.ekispert.com`) under a seeded stop-reward benchmark and compare it with NSGA-II, ILS+2opt, VNS, SA, McRAPTOR, Pure RAPTOR, Greedy, ALNS (adapted), and DynPref (adapted). ReRAPTOR achieves the highest overall average stop score (826.90), improving on NSGA-II by 8.71%, while remaining 6.57× faster on average (4165.67 ms vs 27351.18 ms). An additional balance analysis further shows that ReRAPTOR is the closest method to the score–time ideal point, with query-level ideal distance 0.1672 versus 0.6292 for NSGA-II. Ablation results show that Beam and Dominance are the two core scalability layers, whereas Negative pruning mainly improves semantic alignment in theme-constrained tasks. These results indicate that ReRAPTOR provides a strong quality–speed trade-off for value-oriented transit itinerary planning.

Keywords Transit itinerary planning · Transit schedule data · RAPTOR · Time-budgeted route planning · Multi-objective search

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1 Introduction

Public-transit route planning is traditionally formulated as a timetable-feasible earliest-arrival problem [2,6,8, 14]. That formulation is appropriate for commuters, but it does not capture the needs of leisure travelers, city explorers, or users who want to spend a fixed amount of time on transit while collecting as much route value as possible. In these settings, a good itinerary is not defined only by arrival time. It may also depend on ride duration, the number of distinct stops visited, spatial spread, and the thematic or semantic quality of the route. The problem is therefore closer to time-budgeted

itinerary design than to classical shortest-path search [13,23,24].

The main technical difficulty is the combination of timetable feasibility and value-oriented search. Every candidate route must respect scheduled departures, arrivals, and transfer constraints, while the search objective depends on accumulated route structure rather than on arrival time alone. Once stop diversity, spatial expansion, revisit patterns, or task-specific stop rewards are incorporated, the number of plausible partial routes grows quickly. An overly aggressive compression strategy removes information needed for value maximization, whereas an unpruned search becomes computationally unstable.

This paper addresses that gap with ReRAPTOR, a reverse-objective RAPTOR (Round-bAsed Public Transit Optimized Router)-style search over timetable-feasible transit stop schedules [8,26,27]. Instead of propagating labels to minimize arrival time, ReRAPTOR preserves the timetable-feasible expansion logic of RAPTOR while ranking partial routes by a task-dependent value surrogate. The framework is generic: the search backbone is shared, while the scoring and acceptance modules can be instantiated for geometry-aware exploration or for the seeded stop-reward task used in our main experiments. To keep the search tractable, ReRAPTOR combines three pruning layers: Beam pruning for width control, Dominance pruning for local label filtering, and Negative pruning for rejecting semantically weak or clearly low-quality partial routes, shifting the goal from earliest-arrival routing to time-budgeted value collection.

We evaluate the method on Ekispert-provided transit schedule data for Fukuoka through four experiment groups. Experiment 1 compares ReRAPTOR with NSGA-II, ILS+2opt, VNS, SA, McRAPTOR, Pure RAPTOR, Greedy, and two adapted recent baselines (ALNS, DynPref) under a seeded stop-reward benchmark; ReRAPTOR reaches the best overall average stop score (826.90) and is 6.57× faster than NSGA-II on average (4165.67 ms vs 27351.18 ms). Experiment 2 isolates pruning effects and shows Beam and Dominance are the two core scalability layers. Experiment 3 adds quality–speed balance analysis and Experiment 4 provides route-level interpretability cases, including a counterexample.

The contributions are fourfold: (1) a timetable-feasible formulation with pluggable objective/acceptance modules; (2) ReRAPTOR, which optimizes value during RAPTOR-style expansion rather than post-hoc route scoring; (3) a three-layer pruning analysis showing Beam and Dominance as core scalability controls and Negative pruning as semantic-alignment control; and (4) an empirical study on Ekispert-based Fukuoka schedules showing strong quality–speed balance while explicitly reporting failure cases.

The remainder of the paper reviews related work, formulates the problem, presents the method, and reports experiments, discussion, and conclusions.

2 Related Work

Timetable-based public-transit routing has been studied extensively, with RAPTOR [8] being a representative round-based framework for efficient earliest-arrival and Pareto-feasible journey search. Later work improves transfer handling, bicriterion search, restricted Pareto sets, and acceleration through trip-based, transfer-aware, and partition-based methods [1,2,6,7,14,21]. Related studies also extend public-transport routing toward fare-aware or multi-objective settings [10,25]. In parallel, GTFS and timetable-derived datasets have become standard substrates for transit analysis and reproducible evaluation [12,16,19,26,27]. These directions make timetable-feasible routing practical, but they do not directly address fixed-budget itinerary design where route value depends on accumulated structure and exploration history.

On the itinerary-design side, the tourist trip design problem and the broader orienteering family study budget-constrained profit collection, time windows, and preference-aware route construction [13,23,24]. Representative works consider personalized tourist planning, team orienteering, and route recommendation under user preferences [11,20,22,24]. More generally, multi-objective route search and general-purpose optimizers such as NSGA-II, iterated local search, variable neighborhood search, and simulated annealing provide relevant baselines [4,5,9,15,17,18]. Recent work also includes dynamic preference-based multimodal trip planning and demand-adaptive multi-route planning with ALNS-style search, as well as integrated intermodal timetabling and scheduling [29,30,28]. A recent review further summarizes metaheuristic trends in multimodal transport optimization [3]. Many recent systems assume online inputs (e.g., shared-mobility state or interactive preferences) outside our fixed timetable-only benchmark, so we compare directly compatible methods and include two recent *adapted* baselines under matched constraints. Our contribution differs in that it embeds a pluggable value module directly inside timetable-feasible RAPTOR-style expansion instead of optimizing route value only after constructing complete itineraries.

3 Problem Formulation

We model the transit timetable as a temporal directed graph [19,26,27] $G = (\mathcal{S}, \mathcal{E})$, where each stop $s \in \mathcal{S}$ is a station identifier and each edge $e \in \mathcal{E}$ is a feasible timetable segment $e = (u, v, \tau^{\mathrm{dep}}(e), \tau^{\mathrm{arr}}(e), \ell(e), d(e))$. Here $u, v \in \mathcal{S}$ are the origin and destination stop IDs of a segment, τ^{dep} and τ^{arr} are timetable departure/arrival timestamps, $\ell(e)$ is the transit line identifier, and $d(e)$ is the segment-distance attribute (also written as d_e in the extension equation). Given a query $q = (o, t_0, B)$ with origin o, departure time t_0, and time budget B, the task is to construct a timetable-feasible itinerary $r = (e_1, \ldots, e_m)$ that maximizes route value.

A route is feasible if it starts from the query origin, respects stop continuity, satisfies transfer-buffer constraints, and finishes within the budget:

$$\begin{aligned} u(e_1) &= o, \\ \tau^{\mathrm{dep}}(e_1) &\geq t_0, \\ v(e_i) &= u(e_{i+1}), \\ \tau^{\mathrm{dep}}(e_{i+1}) &\geq \tau^{\mathrm{arr}}(e_i) + \delta(e_i, e_{i+1}), \\ \tau^{\mathrm{arr}}(e_m) &\leq t_0 + B, \end{aligned} \tag{1}$$

where $\delta(e_i, e_{i+1}) = 0$ for same-trip continuation and $\delta(e_i, e_{i+1}) = \Delta_{\mathrm{tr}}$ otherwise.

Instead of fixing one route-value function for all tasks, we define a generic objective module parameterized by θ:

$$\max_{r \in \mathcal{R}(q)} F_\theta(r) \qquad \text{s.t. } g_\theta(r) = 1, \tag{2}$$

where $\mathcal{R}(q)$ is the set of timetable-feasible routes under Eq. 1, F_θ is a scenario-specific route-value function, and g_θ is an optional acceptance predicate. This separation is important in our setting because the search backbone remains unchanged across experiments, while only the value module and acceptance logic are instantiated differently.

The default geometry-aware instance in the original formulation evaluates four positive criteria: $f_1(r) =$ ride duration, $f_2(r) =$ number of unique visited stops, $f_3(r) =$ convex-hull area of visited stop coordinates, and $f_4(r) =$ quadrant coverage around the city center. Using normalizers $c_1 = 480$, $c_2 = 50$, $c_3 = 100$, and $c_4 = 4$, we define $\tilde{f}_j(r) = f_j(r)/c_j$ and optimize

$$F_{\mathrm{geo}}(r) = \sum_{j=1}^{4} \alpha_j \tilde{f}_j(r), \tag{3}$$

with default weights $(0.40, 0.25, 0.20, 0.15)$ from the original Fukuoka geometry-aware system.

The seeded stop-reward instantiation used in Experiments 1–2 and reused for the balance analysis in Experiment 3 assigns each stop $s \in \mathcal{S}$ a reward $w_\omega(s)$ drawn once from a random seed ω, and each unique non-origin stop incurs a dwell time η. Let $V^+(r)$ denote the set of unique visited stops excluding the origin, and let $T(r)$ be the elapsed timetable travel time of route r. The effective task time is

$$T_\eta(r) = T(r) + \eta\,|V^+(r)|. \tag{4}$$

The seeded stop-reward objective is then

$$F_{\omega,\eta}(r) = \sum_{s \in V^+(r)} w_\omega(s), \qquad g_{\omega,\eta}(r) = \mathbf{1}[T_\eta(r) \leq B]\,. \tag{5}$$

In other words, the main experiments no longer optimize hull area directly; they optimize collected stop reward under a fixed effective-time budget, while the same timetable-feasible search structure and pruning framework are retained. We use $F_\theta(\cdot)$ for the final evaluated route objective and $S_\theta(\cdot)$ only for ranking partial labels during search.

4 Proposed Method

4.1 Overall Framework

ReRAPTOR is a RAPTOR-style label-setting search over timetable-feasible transit stop schedules with pluggable objective modules and a three-layer pruning mechanism [8,14]. Unlike earliest-arrival transit routing, the search state explicitly stores route history and exploration statistics. Partial itineraries can therefore be ranked by both budgeted utility and route structure. At a high level, the planner initializes an origin label, repeatedly expands timetable-feasible outgoing edges from marked stops, scores each partial itinerary with an objective-specific surrogate, prunes it with the three-layer filter, and returns the highest-scored accepted route.

Figure 1 highlights the separation between the shared search backbone and the pluggable scoring/acceptance modules.

4.2 Core Algorithm

Each partial route is represented by a label

$$\lambda = \left(a, \rho, d, V, q, \mathit{\Pi}, c^{\mathrm{stop}}, c^{\mathrm{line}}, n^{\mathrm{tr}}, \psi\right), \tag{6}$$

where a is the current arrival time, ρ is accumulated ride time, d is traveled distance, V is the set of visited stops, q is a bit-mask encoding covered quadrants, $\mathit{\Pi}$ is the ordered leg sequence, c^{stop} and c^{line} are visit counters,

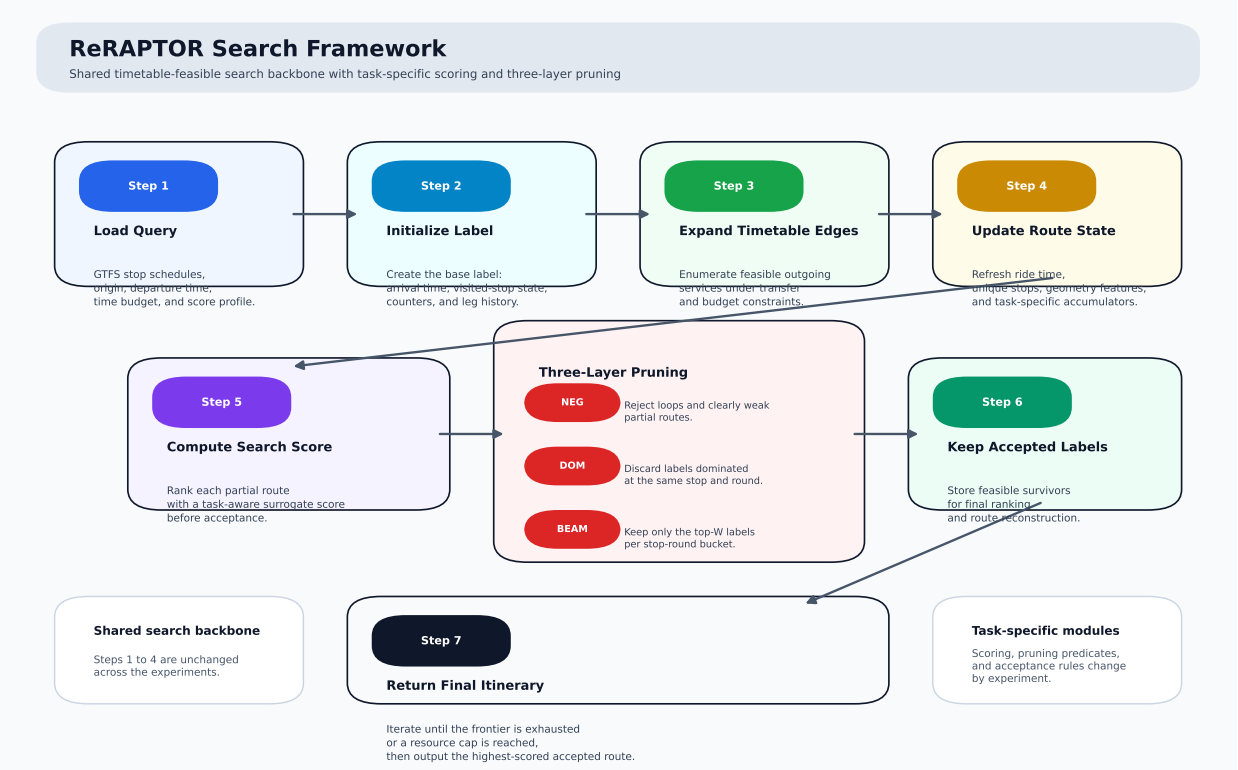


Fig. 1 Overall framework of the proposed time-budgeted itinerary planner. The core search remains unchanged across experiments; only score weights or optional acceptance predicates are varied.

and n^{tr} is the number of transfers. The additional term ψ stores objective-specific accumulators. In the seeded stop-reward task used in Experiments 1–2 and reused in Experiment 3, ψ includes at least the collected reward total σ and the number of unique scored stops m.

Given a feasible edge $e = (u, v, \tau^{\mathrm{dep}}, \tau^{\mathrm{arr}}, \ell, d_e)$, the extension operator updates the label by

$$\begin{aligned} a' &= \tau^{\mathrm{arr}}(e), \\ \rho' &= \rho + \left(\tau^{\mathrm{arr}}(e) - \tau^{\mathrm{dep}}(e)\right), \\ d' &= d + d_e, \qquad V' = V \cup \{v\}, \\ q' &= q \vee q(v), \qquad \Pi' = \Pi \oplus e, \\ \psi' &= U_\theta(\psi, \lambda, e), \end{aligned} \tag{7}$$

while the stop and line counters are incremented on the destination stop and on new line boardings, respectively. Here $q(v)$ denotes the quadrant indicator of stop v, and U_θ is an objective-specific state update. For the stop-reward instantiation, U_θ adds $w_\omega(v)$ only when v is a newly scored stop and increments m accordingly. This state design makes geometric coverage, revisit patterns, transfer behavior, and task-specific reward accumulation available during search rather than only after route construction.

The search surrogate used for label ranking is written in generic form as

$$S_\theta(\lambda) = \Phi_\theta(\lambda, \psi(\lambda)) - P(\lambda), \tag{8}$$

where Φ_θ is the task-specific positive utility module and $P(\lambda)$ is the negative-pruning term. The negative penalty is

$$\begin{aligned} P(\lambda) = 50000\,\mathbf{1}\Big[\max_s c_\lambda^{\mathrm{stop}}(s) > 2\Big] \\ + 20000\,\mathbf{1}[\rho(\lambda)/|V(\lambda)| < 2] \\ + P_\theta(\lambda), \end{aligned} \tag{9}$$

where P_θ is an optional scenario-specific penalty used only when theme or scenario constraints are explicitly instantiated. The threshold $\max_s c_\lambda^{\mathrm{stop}}(s) > 2$ suppresses clear revisit loops, while $\rho(\lambda)/|V(\lambda)| < 2$ suppresses chains of unrealistically short hops; the coefficients 50000 and 20000 are safety-side heuristic defaults inherited from the early implementation, not values tuned from the reported experiments.

For the geometry-aware default used in the original system, the positive utility term Φ_θ favors ride time, stop diversity, and spatial coverage with fixed heuristic coefficients.

For the seeded stop-reward task used in Experiments 1–2 and reused in Experiment 3, the search score

is lexicographically biased toward collected reward and only uses time-based terms as tie-breakers:

$$S_{\omega,\eta}(\lambda) = 10000\,\sigma(\lambda) + 5\ \mathrm{SPH}(\lambda) - 0.1\,T_\eta(\lambda) - 50000\ \max(0, T_\eta(\lambda) - B) - P(\lambda), \tag{10}$$

where $\sigma(\lambda)$ is the collected stop-reward total, $T_\eta(\lambda)$ is the effective time of the partial route, and $\mathrm{SPH}(\lambda) = 60\,\sigma(\lambda)/T_\eta(\lambda)$ is the score-per-hour tie-breaker. The large coefficient on $\sigma(\lambda)$ ensures that even a small gain in collected reward dominates secondary terms, which matches the seeded stop-reward evaluation protocol. The constants $10000, 5, 0.1$, and 50000 are design defaults for implementing the intended lexicographic priority "reward $\gg$ efficiency $\gg$ time"; the factor 60 in SPH is only a minute-to-hour unit conversion. For clarity, F_θ in Eq. 2 evaluates a complete route, while S_θ in Eq. 8 is a surrogate used to rank partial labels during search.

After scoring, each candidate is filtered by two additional pruning stages. First, dominance pruning discards labels that are clearly weaker than an existing label at the same stop and round. As in multi-objective shortest-path and multicriteria transit-routing literature, this step removes labels that are unlikely to produce better final outcomes [9,14]. In generic form, λ^a dominates λ^b if

$$\lambda^a \succeq_\theta \lambda^b \iff D_\theta(\lambda^a, \lambda^b) = 1. \tag{11}$$

Here $D_\theta(\cdot,\cdot)$ denotes the task-specific dominance predicate. For the geometry-aware default, dominance is decided heuristically from search score, ride time, and spatial coverage. For the stop-reward task, we use a tighter seeded stop-reward rule based on reward, effective time, and an optimistic upper bound:

$$\begin{aligned}\lambda^a \succeq_{\omega,\eta} \lambda^b \iff\ & \sigma(\lambda^a) \geq \sigma(\lambda^b) \\ & \wedge\ T_\eta(\lambda^a) \leq T_\eta(\lambda^b) \\ & \wedge\ U_{\omega,\eta}(\lambda^a) \geq U_{\omega,\eta}(\lambda^b),\end{aligned} \tag{12}$$

with at least one strict inequality. Here $U_{\omega,\eta}(\lambda)$ is an optimistic upper bound on the final stop reward, computed from the current reward total plus the best uncollected stop scores that can still fit into the remaining effective-time budget. Second, beam pruning keeps only the best W labels in each stop-round bucket:

$$\begin{aligned}\mathcal{L}_{k+1}(v) = \mathrm{TopW}_{S_\theta}\Big(& \big\{\lambda' \,\big|\, \lambda \in \mathcal{L}_k(u), \\ & e \in \mathrm{Next}(u, a_\lambda), \\ & \lambda' = \mathrm{Extend}(\lambda, e)\big\}\Big).\end{aligned} \tag{13}$$

Here TopW_{S_θ} returns the highest-scored W labels under S_θ, i.e., candidates are scored by S_θ, sorted in descending order, and truncated to at most W labels (if fewer than W candidates exist, all are kept). $\mathrm{Next}(u, a_\lambda)$ denotes the feasible outgoing timetable edges from stop u no earlier than arrival time a_λ. Here k is the round index and $\mathcal{L}_k(v)$ is the label set stored at stop v after round k. In our experiments, $W = 50$ is the default setting; this is a fixed practical beam-width preset rather than a value optimized in the paper. The route returned at the end of search is the highest-scored accepted label, where acceptance requires at least two legs and may optionally include an experiment-specific predicate.

4.3 Complexity and Implementation

With beam width W, each stop-round bucket stores at most W labels, so practical runtime is bounded by the number of marked stops, feasible outgoing edges, and bounded per-bucket dominance and beam maintenance; memory is $O(|\mathcal{S}|W)$ per round. The implementation directly expands stop schedules through `_next_edges(...)` rather than reconstructing route tables because some `trip_id` values are not globally unique in our dataset. It memoizes geometry tuples by label signature, uses a 120-minute lookahead window, and sets $R = \min(220, \max(40, \lfloor B/2 \rfloor + 20))$ with timeout and label-cap guards as fixed practical defaults. Here R is the maximum number of search rounds (timetable-expansion iterations), and B is the query time budget in minutes. The inner term $\lfloor B/2 \rfloor + 20$ scales search depth with budget length, while the outer min/max bounds keep R in $[40, 220]$.

5 Experiments

5.1 Experiment 1: Comprehensive Algorithm Comparison

5.1.1 Objective

Experiment 1 evaluates whether ReRAPTOR provides better overall route quality than representative baselines after reformulating the benchmark to a seeded stop-reward task. Instead of relying on hull area as the primary benchmark outcome, each stop is assigned a random score and the task is to maximize the collected stop score within a fixed time budget. The three evaluation questions are:

- **RQ1 (quality):** Does ReRAPTOR improve the primary task metric, `stop_score_total`?
- **RQ2 (efficiency):** Does ReRAPTOR achieve high score while remaining fast on many queries?

Table 1 Overall summary of Experiment 1 (query-level means, 120 queries).

Method	Stop Score	Elapsed (ms)	Score / hour	Avg. rank
Ours (Full)	826.90	4165.67	193.02	2.12
NSGA-II	760.66	27351.18	170.40	2.94
ILS+2opt	697.46	518.79	157.10	3.53
ALNS (adapted)	657.76	62.83	150.96	4.55
VNS	645.00	169.31	148.13	4.88
McRAPTOR	546.70	3684.86	149.33	6.12
SA	589.46	130.80	137.25	6.38
Pure RAPTOR	320.31	56483.31	192.36	7.78
DynPref (adapted)	443.70	10.91	111.16	8.03
Greedy (L=3)	416.99	3.05	101.27	8.66

– **RQ3 (robustness):** Are the gains stable across budgets and random stop-score seeds?

5.1.2 Setup and Protocol

Baseline selection follows three criteria: task compatibility with timetable-feasible fixed-budget routing, reproducible implementation in our pipeline, and diversity across optimizer families. We therefore compare general-purpose metaheuristics, transit-routing baselines, and two recent adapted methods (ALNS, DynPref). Methods requiring external online signals unavailable in this benchmark are not used as direct one-to-one baselines.

All compared methods share the same Fukuoka timetable dataset (304,344 timetable segments, 1,032 stops, and 302 lines), the same query grid of 10 origins × 3 departure times (`07:00`, `10:00`, `14:00`) × 4 budgets (`2h`, `4h`, `6h`, `8h`), and the same five stop-score seeds per query. The comparison includes Ours, NSGA-II [5], ILS+2opt [17], VNS [18], SA [15], McRAPTOR [8], Pure RAPTOR [8], Greedy, ALNS (adapted) [30], and DynPref (adapted) [29]. For fairness, all methods (including adapted baselines) use the same timetable inputs, stop-score map, dwell-time rule, budget-feasibility check, and query protocol; adapted variants only replace incompatible components. This yields 600 task rows per algorithm and 120 query-level means after averaging over score seeds. For each task key (origin, depart_time , budget, score_seed), all methods are evaluated under the same stop-score map and dwell-time setting.

5.1.3 Main Results

Table 1 gives the overall (2h+4h+6h+8h) query-level means. ReRAPTOR achieves the highest average stop score (826.90), exceeding NSGA-II (760.66) by +8.71%, while also being 6.57× faster overall (4165.67 ms vs 27351.18 ms). The main claim of Experiment 1 is therefore not only "best average score", but more importantly that Ours achieves *high score and fast response on many queries*.

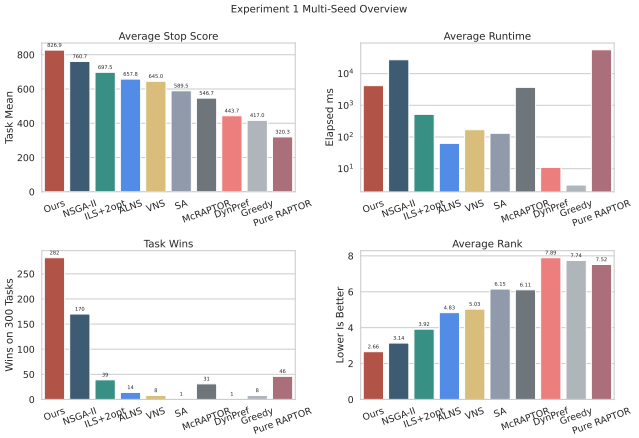


Fig. 2 Overview of Experiment 1 under the score-guided multi-seed benchmark. The central message is that Ours most often lies in the high-score and low-runtime region, while also ranking first in average stop score.

In Table 1, *Stop Score* is the query-level mean of `stop_score_total` (i.e., $F_{\omega,\eta}(r)$ in Eq. 5); *Elapsed (ms)* is query-level mean wall-clock generation time; *Score / hour* is SPH $= F_{\omega,\eta}(r)/(T_\eta(r)/60)$; and *Avg. rank* orders methods by higher score then lower elapsed time.

At the task level, Ours beats NSGA-II in 368/600 comparisons. After averaging over stop-score seeds, Ours still achieves a higher stop score than NSGA-II in 82/120 base queries. Figure 2 visualizes this combined view and summarizes the main result of Experiment 1.

5.1.4 Budget-Sliced Analysis

To avoid mixed-budget interpretation bias, we report the 2h, 4h, 6h, and 8h settings separately. Ours remains stronger than NSGA-II in both settings:

– 2h: stop score 350.37 vs 322.62, while runtime is 83.95 ms vs 17026.77 ms.
– 4h: stop score 725.78 vs 624.94, while runtime is 1329.04 ms vs 26629.94 ms.
– 6h: stop score 973.57 vs 900.64, while runtime is 2226.92 ms vs 29981.78 ms.
– 8h: stop score 1257.89 vs 1194.46, while runtime is 13022.77 ms vs 35766.26 ms.

At the task level, Ours outperforms NSGA-II in 69/150, 113/150, 90/150, and 96/150 comparisons for 2h, 4h, 6h, and 8h, respectively. Figure 3 visualizes this budget-sliced progression.

5.1.5 Robustness Across Score Seeds

The score gap stays positive for all tested seeds. Across the five seeds, the average Ours–NSGA-II gap ranges from +15.15 to +107.43, and the observed zero-score

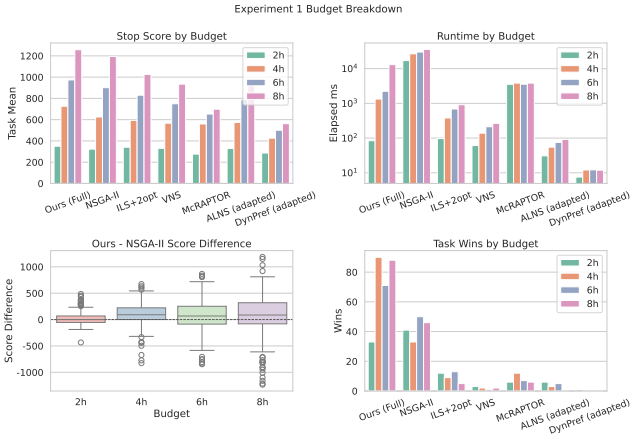


Fig. 3 Budget-sliced overview of Experiment 1. The overall view highlights the global score–runtime pattern, whereas the budget breakdown makes the 2h-to-8h progression explicit.

cases of Ours are `effective_budget_feasible = False` edge cases rather than valid-but-poor routes. Concretely, this means the effective-time constraint $T_\eta(r) \leq B$ is not satisfiable for that specific (origin , depart_time, budget, score_seed) task, so the case is treated as infeasible, not as a feasible task where Ours returns an abnormally weak route.

5.2 Experiment 2: Three-Layer Pruning Ablation and Supporting Evidence

5.2.1 Objective and Research Questions

Experiment 2 isolates the three pruning layers in ReRAPTOR: *Beam pruning* for width control, *Dominance pruning* for local label filtering, and *Negative-score pruning* for rule- or theme-aligned rejection. We answer the following questions:

- **RQ4 (core scalability):** Are Beam and Dominance jointly indispensable for tractable search?
- **RQ5 (budget sensitivity):** How do Beam and Dominance behave at 4h and 8h?
- **RQ6 (semantic alignment):** Does Negative pruning improve route-theme preservation in project-style theme tasks?

5.2.2 Main Ablation Protocol

The main ablation uses the same base query grid as Experiment 1: 10 origins × 3 departure times × 2 budgets $(4\text{h}, 8\text{h})$ × 3 stop-score seeds. With 8 algorithmic configurations, this yields 1440 runs in total.

The evaluated variants are `Full`, `NoNeg`, `NoBeam`, `NoDom`, `NoBeam-NoDom`, `NoDom-NoNeg`, `NoBeam-NoNeg`, and `NoBeam-NoDom-NoNeg`. The main metrics are stop score, runtime, label count, and cap-hit rate.

Table 2 Main ablation summary of Experiment 2 (task-level means over 180 task units).

Method	StopScore	Elapsed (ms)	Labels	CapHit (%)
Ours (Full)	999.27	4668.15	115.4K	0.00
Ours (NoNeg)	996.10	4865.37	119.3K	0.00
Ours (NoDom)	971.91	13589.45	349.9K	0.00
Ours (NoBeam)	912.04	44486.93	1.49M	59.44
Ours (NoBeam-NoDom)	313.79	61694.58	2.81M	100.00
Ours (NoDom-NoNeg)	973.06	12984.17	351.0K	0.00
Ours (NoBeam-NoNeg)	911.56	44700.47	1.51M	60.00
Ours (NoBeam-NoDom-NoNeg)	289.37	20929.99	1.00M	100.00

Table 2 shows that `Ours (Full)` has the best combined profile across score, runtime, and search size. `Ours (NoNeg)` is nearly tied on score, which is consistent with the claim that Negative pruning is not the main speedup layer in the generic benchmark. By contrast, removing either Dominance or Beam is clearly expensive. Relative to Full, `NoDom` is 2.91× slower and uses 3.03× more labels, while `NoBeam` is 9.53× slower and uses 12.89× more labels. The newly added `NoBeam-NoDom` variant is the key evidence: once the two main search-space control layers are removed together, the score drops by −685.48, runtime grows to 13.22× Full, and label count grows to 24.36× Full.

Therefore, the updated interpretation of Experiment 2 is that **Beam and Dominance are co-core efficiency layers**. Negative pruning has a different role and should be argued mainly through supporting tasks.

Supplementary analyses are consistent with the same role separation. On the 4h/8h budget slices, removing Beam can buy some extra score at 4h (803.80 vs 737.04) only at the cost of 35.10× runtime and cap hits, whereas at 8h `Full` dominates `NoBeam` on both score and runtime and `NoBeam` hits the cap on all tasks. On a supplementary project-style theme-constrained route study with 384 paired `Full` vs `NoNeg` cases under identical starts, departures, and budgets, `Full` improves ThemeHit from 0.677 to 0.965 and ThemeSat from 0.646 to 0.958, where ThemeHit denotes the proportion of theme-matched visited stops and ThemeSat denotes route-level theme satisfaction, confirming that Negative pruning mainly improves objective alignment rather than generic stop-score efficiency.

5.2.3 Interpretation: Why the Three Layers Are All Meaningful

Experiment 2 provides complementary evidence:

- **Dominance pruning** is one of the two core scalability layers; removing it causes a clear search-space blow-up.
- **Beam pruning** is the other core scalability layer; removing it can occasionally buy score at short bud-

gets, but it becomes unstable and expensive as the budget grows.
– **Negative pruning** is a semantic-alignment layer; it improves route-theme preservation, with an explicit runtime trade-off.

Therefore, the three-layer design is justified as a *functional decomposition*: joint search-space control (Beam + Dominance) and objective alignment (Negative).

5.3 Experiment 3: Balance Analysis Supplementing Experiment 1

5.3.1 Objective and Protocol

Experiment 3 is an additional balance analysis derived from the Experiment 1 source results. Using the same 120 base queries after averaging over stop-score seeds, it asks whether the methods remain close to the score–time ideal point under the same seeded stop-reward task.

5.3.2 Main Results

ReRAPTOR is the closest method to the score–time ideal point. Its query-level ideal distance is 0.1672, compared with 0.6292 for NSGA-II, and it is simultaneously higher-score and faster in 82/120 base queries. The budget-sliced ideal distances of ReRAPTOR are 0.2482, 0.1097, 0.1656, and 0.3342 for the 2h, 4h, 6h, and 8h settings, respectively, versus 0.7026, 0.5888, 0.5963, and 0.6602 for NSGA-II. Therefore, Experiment 3 supports the same interpretation as Table 1: the main strength of ReRAPTOR is not only a higher average stop score, but a stronger overall quality–speed balance.

5.4 Experiment 4: Route Case Study

5.4.1 Objective and Positioning

Experiment 4 is not designed as another benchmark. Its purpose is to illustrate the main findings of Experiments 1 and 3 through concrete route examples, making the results more interpretable. We address the following questions:

– **RQ10 (success case):** Can the score and speed advantages of Ours be visually confirmed on specific queries?
– **RQ11 (balance case):** Is the quality–speed balance claim reflected in the route structure?
– **RQ12 (limitation):** Are there queries where Ours does not achieve the highest score?

5.4.2 Case Selection Protocol

To avoid arbitrary cherry-picking, the cases are selected from the Experiment 1 results using systematic criteria. We use three cases from the score-guided benchmark:

– **Case A (success):** the query where Ours clearly wins on both stop score and generation time;
– **Case B (balance):** a query where Ours is not the top scorer, but achieves the best quality–speed balance;
– **Case C (counterexample):** a query where Ours is faster but NSGA-II reaches a higher score.

Each case compares Ours with NSGA-II in the main text, while the full three-way numeric breakdown, including score per hour and scored stops, is deferred to the supplementary materials.

5.4.3 Main Results

Case A is the clearest success example. Ours achieves a stop score of 1459.0 with a generation time of 9091.1 ms, while NSGA-II reaches only 854.0 in 14129.0 ms. This confirms that there exist queries where Ours simultaneously achieves a substantially higher score and a shorter generation time.

Case B is the intended quality–speed balance example. Here Ours does not achieve the highest score (1276.0 vs 1590.0 for NSGA-II), but its generation time is much shorter (4473.7 ms vs 18956.2 ms). This case visually reinforces the Experiment 3 claim that Ours most often stays close to the score–time ideal point. Figure 4 visualizes the main route-level contrast between the success case and the balance case.

Case C is kept as a counterexample. Ours is faster (2730.8 ms vs 20367.1 ms) but does not reach the highest score (1358.0 vs 1556.0 for NSGA-II). The detailed three-case numeric table, including the counterexample route, is deferred to the supplementary materials so that the main paper can keep the route-level visual comparison without duplicating the same information twice.

5.4.4 Interpretation

Experiment 4 provides a route-level interpretability check for the aggregate findings. Case A and Case B visually confirm the quantitative claims of Experiments 1 and 3, while Case C improves credibility by showing a limitation.

6 Discussion

The experimental results support a more precise claim than simply “ReRAPTOR obtains the best routes.” Un-

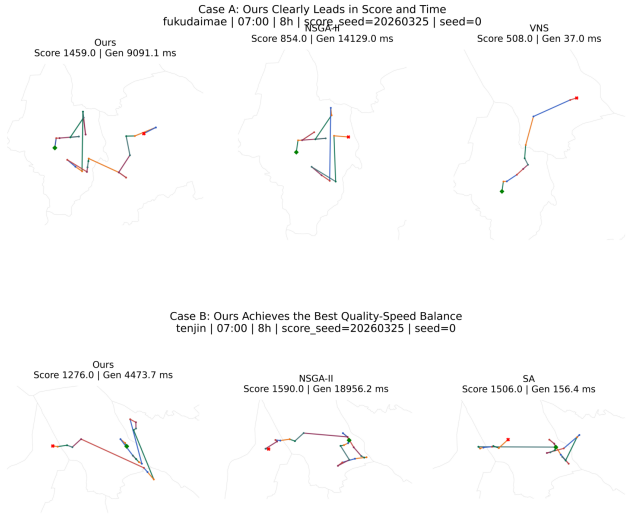


Fig. 4 Route case study panels. Top (Case A): Ours is higher-score and faster. Bottom (Case B): Ours is not top score but has better quality–speed balance.

der the seeded stop-reward benchmark, its key strength is the consistency of the quality–speed balance. In Experiment 1, ReRAPTOR achieves the highest average stop score, improves on NSGA-II by +8.71%, and remains 6.57× faster on average. Experiment 3 is consistent: ReRAPTOR has the best query-level ideal distance (0.1672) and is simultaneously higher-score and faster than NSGA-II in 82/120 base queries. Experiment 4 also shows that it does not win every individual query, improving the credibility of the evaluation.

The ablation results also clarify how the proposed method should be understood. Beam and Dominance are the two core scalability controls for keeping label search tractable. Removing Dominance substantially increases runtime and label count; removing Beam is even more expensive, especially at the 8-hour budget where cap hits occur in all tasks. Removing both leads to rapid search-space growth and lower route quality. Negative pruning has a different role: it is not the main speedup factor on the generic stop-reward benchmark, but it improves objective alignment in the theme-constrained study. The evaluation is still limited to one city's Ekispert schedule data and a seeded reward proxy instead of learned user preferences; future work should study cross-city transfer, learned reward modules, and adaptive pruning.

7 Conclusion

This paper presented ReRAPTOR, a reverse-objective RAPTOR-style method for time-budgeted value-maximizing transit itinerary planning on timetable-based transit schedules. Its main methodological contribution is to optimize route value during timetable-feasible expansion through a shared search backbone with pluggable objective modules and a three-layer pruning design, rather than scoring candidate routes only after they are fully constructed.

Across the main stop-reward benchmark, ReRAPTOR achieves the best overall quality–speed profile among the compared baselines. It attains the highest average stop score (826.90), exceeds NSGA-II by 8.71% on that metric, and remains 6.57× faster on average. Experiment 3 shows that it is also the closest method to the score–time ideal point, and the ablation study shows that Beam and Dominance are the two core scalability layers, whereas Negative pruning mainly improves semantic alignment in constrained theme tasks. The route case study further confirms both the strengths of the method and the fact that it does not win every query.

Overall, ReRAPTOR is a practical approach for value-oriented transit itinerary planning when the goal is high route value with stable response time.

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Supplementary Materials

Additional tables, source data files, and supplementary images will be released at `https://github.com/Akihito-zhou/reraptor_paper`.

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